CYBERSRS

Secure Vulnerability Remediation System

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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1. Introduction

Original project description

Design a secure vulnerability-remediation tracking system for a university that manages findings from approved security assessment reports for servers, applications, endpoints, and network devices. Security analysts should be able to review findings, assign remediation tasks to authorized departmental administrators, record investigation notes, track remediation status, and close findings after reviewing submitted evidence. The system should restrict access according to user roles, detect unauthorized changes to findings or remediation records, generate security alerts, and maintain audit logs of finding updates and administrator actions. The platform should provide a centralized dashboard, integrate with existing university identity services, support multiple campus locations, and remain secure, reliable, scalable, and manageable.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder source statement E-001

Detail: Design a secure vulnerability-remediation tracking system for a university that manages findings from approved security assessment reports for servers, applications, endpoints, and network devices.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: Security analysts should be able to review findings, assign remediation tasks to authorized departmental administrators, record investigation notes, track remediation status, and close findings after reviewing submitted evidence.

Status: confirmed

Evidence ids: E-002

Stakeholder source statement E-003

Detail: The system should restrict access according to user roles, detect unauthorized changes to findings or remediation records, generate security alerts, and maintain audit logs of finding updates and administrator actions.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: The platform should provide a centralized dashboard, integrate with existing university identity services, support multiple campus locations, and remain secure, reliable, scalable, and manageable.

Status: confirmed

Evidence ids: E-004

2. Functional requirements

Applicability reason: Independent source obligations; unspecified choices remain TBD.

Requirements

FR-001 - manage findings from approved security assessment reports for servers, applications, endpoints, and network devices

Statement: The system shall manage findings from approved security assessment reports for servers, applications, endpoints, and network devices.

Rationale: Source obligation E-001_O01.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: the system receives an approved security assessment report for a server

When: the system manages the findings from that report

Then: the system returns a structured finding list with severity, location, and recommended remediation steps

Method: test

Observable evidence: External test capture of the resulting response

FR-002 - allow Security analysts to review findings

Statement: The system shall allow Security analysts to review findings.

Rationale: Source obligation E-002_O01.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: Security analysts review findings named in the request

When: Security analysts initiate the review action

Then: The requested findings are returned for examination; an external test capture records the response

Method: test

Observable evidence: External test capture of the resulting response

FR-003 - allow Security analysts to assign remediation tasks to authorized departmental administrators

Statement: The system shall allow Security analysts to assign remediation tasks to authorized departmental administrators.

Rationale: Source obligation E-002_O02.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: Security analysts assign remediation tasks to authorized departmental administrators

When: Security analysts initiate the assignment operation

Then: The permission or access state for authorized departmental administrators is observed as active after Security analysts initiate the assignment operation.

Method: inspection

Observable evidence: External test capture of the resulting state

FR-004 - allow Security analysts to record investigation notes

Statement: The system shall allow Security analysts to record investigation notes.

Rationale: Source obligation E-002_O03.

Priority: should

Dimensions: business; data

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: Security analysts perform a controlled instance of recording investigation notes

When: Security analysts initiate the record operation

Then: The requested investigation note record is available for inspection; an external test capture records the inspection record

Method: analysis

Observable evidence: External test capture of the inspection record

FR-005 - allow Security analysts to track remediation status

Statement: The system shall allow Security analysts to track remediation status.

Rationale: Source obligation E-002_O04.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: Security analysts track remediation status for a specific finding

When: Security analysts initiate the tracking operation

Then: The named history of remediation status is available for examination; an external test capture records the resulting state

Method: demonstration

Observable evidence: External test capture of the resulting state

FR-006 - allow Security analysts to close findings after reviewing submitted evidence

Statement: The system shall allow Security analysts to close findings after reviewing submitted evidence.

Rationale: Source obligation E-002_O05.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: Security analysts close findings after reviewing submitted evidence

When: Security analysts initiate the closure action

Then: The finding is marked as closed; an external test capture records the updated status response

Method: test

Observable evidence: External test capture of the resulting response

FR-007 - restrict access according to user roles

Statement: The system shall restrict access according to user roles.

Rationale: Source obligation E-003_O01.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: system restricts access to resources based on user roles

When: the system enforces role-based access control during resource retrieval

Then: The system enforces role-based access control during resource retrieval, and access to resources is restricted according to user roles.

Method: test

Observable evidence: External test capture of the resulting response

FR-008 - detect unauthorized changes to findings or remediation records

Statement: The system shall detect unauthorized changes to findings or remediation records.

Rationale: Source obligation E-003_O02.

Priority: should

Dimensions: business; data

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: system monitors changes to findings or remediation records

When: an unauthorized modification is attempted on a finding or remediation record

Then: the system detects and logs the change as unauthorized

Method: demonstration

Observable evidence: External test capture of the resulting state

Unresolved decisions: Decision Q-dad9a79bf709 is TBD.

FR-009 - generate security alerts

Statement: The system shall generate security alerts.

Rationale: Source obligation E-003_O03.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: system generates security alerts under defined conditions

When: a predefined alert condition is initiated by a threat or anomaly event

Then: a security alert exists in the designated alerting system with required fields populated

Method: test

Observable evidence: External test capture of the resulting response

FR-010 - maintain audit logs of finding updates and administrator actions

Statement: The system shall maintain audit logs of finding updates and administrator actions.

Rationale: Source obligation E-003_O04.

Priority: should

Dimensions: business; auditability; data

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: system maintains audit logs for finding updates and administrator actions

When: an update to a finding or an administrative action is performed

Then: The system maintains audit logs for finding updates and administrator actions, as evidenced by external test capture of the inspection record.

Method: inspection

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-3aae42e94793 is TBD.; Decision Q-64ef2e28335c is TBD.

FR-011 - provide a centralized dashboard

Statement: The system shall provide a centralized dashboard.

Rationale: Source obligation E-004_O01.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: a controlled user opens the source-named dashboard

When: the system provides a centralized dashboard

Then: the dashboard is displayed or available for use

Method: test

Observable evidence: External test capture of the resulting response

FR-012 - integrate with existing university identity services

Statement: The system shall integrate with existing university identity services.

Rationale: Source obligation E-004_O02.

Priority: should

Dimensions: business; integration

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: a controlled user initiates an identity service exchange with the university identity services

When: the system integrates with existing university identity services

Then: interoperation with the external service is observed

Method: demonstration

Observable evidence: External test capture of the resulting state

FR-013 - support multiple campus locations

Statement: The system shall support multiple campus locations.

Rationale: Source obligation E-004_O03.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: a controlled user initiates use from each stated campus location

When: the system supports multiple campus locations

Then: use succeeds at each location

Method: test

Observable evidence: External test capture of the inspection record

3. Non-functional requirements

Applicability reason: Independent source obligations; unspecified choices remain TBD.

Requirements

NFR-001 - be secure

Statement: The system shall be secure.

Rationale: Source obligation E-001_O02.

Priority: must

Dimensions: security

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: A draft acceptance plan for this quality goal: the system shall be secure against known attack vectors and maintain confidentiality, integrity, and availability

When: An analyst inspects or reviews that acceptance plan to determine whether measurable targets are defined

Then: The plan's measurable targets remain TBD or unconfirmed

Method: inspection

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-7b2e5f89a0b1 is TBD.

NFR-002 - remain secure

Statement: The system shall remain secure.

Rationale: Source obligation E-004_O04.

Priority: should

Dimensions: security

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: The system shall remain secure during vulnerability remediation operations.

When: An analyst inspects or reviews that acceptance plan.

Then: The plan's measurable targets remain TBD or unconfirmed.

Method: inspection

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-7b2e5f89a0b1 is TBD.

NFR-003 - remain reliable

Statement: The system shall remain reliable.

Rationale: Source obligation E-004_O05.

Priority: should

Dimensions: availability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: The system shall remain reliable during vulnerability remediation operations.

When: An analyst inspects or reviews that acceptance plan.

Then: The plan's measurable targets remain TBD or unconfirmed.

Method: inspection

Observable evidence: External test capture of the resulting state

Unresolved decisions: Decision Q-819bb986f419 is TBD.

NFR-004 - remain scalable

Statement: The system shall remain scalable.

Rationale: Source obligation E-004_O06.

Priority: should

Dimensions: performance

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: The system shall remain scalable during vulnerability remediation operations.

When: An analyst inspects or reviews that acceptance plan.

Then: The plan's measurable targets remain TBD or unconfirmed.

Method: analysis

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-543394346815 is TBD.

NFR-005 - remain manageable

Statement: The system shall remain manageable.

Rationale: Source obligation E-004_O07.

Priority: should

Dimensions: usability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: The system shall remain manageable during vulnerability remediation operations.

When: An analyst inspects or reviews that acceptance plan.

Then: The plan's measurable targets remain TBD or unconfirmed.

Method: inspection

Observable evidence: External test capture of the resulting state

Unresolved decisions: Decision Q-5268b96e41b0 is TBD.

Related requirement ids: FR-001; FR-011; FR-012; FR-013

4. Security, privacy and compliance

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-007; FR-008; FR-009; FR-010; NFR-001; NFR-002

5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-004; FR-008; FR-010

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001

7. Compliance and policy applicability matrix

Open decisions: Decision Q-66cf5513b374 is TBD.; Decision Q-dad9a79bf709 is TBD.

8. Architecture

Summary: Proposed logical components and interactions mapped to canonical source obligations.

Items

Actor: Security analysts

Detail: Proposed logical endpoint for the source-stated actor: Security analysts.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006

Component name: Actor: Security analysts

Responsibility: Represent the source-stated actor: Security analysts.

Finding Management Module

Detail: Proposed logical owner for FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-002; E-004

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Component name: Finding Management Module

Responsibility: The system shall manage findings from approved security assessment reports for servers, applications, endpoints, and network devices. The system shall allow Security analysts to review findings. The system shall allow Security analysts to assign remediation tasks to authorized

departmental administrators. The system shall allow Security analysts to record investigation notes. The system shall allow Security analysts to track remediation status. The system shall allow Security analysts to close findings after reviewing submitted evidence. The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Access Control Engine

Detail: Proposed logical owner for FR-007, FR-008, FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-003; E-004

Requirement ids: FR-007; FR-008; FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Component name: Access Control Engine

Responsibility: The system shall restrict access according to user roles. The system shall detect unauthorized changes to findings or remediation records. The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Security Alert and Audit Module

Detail: Proposed logical owner for FR-009, FR-010, FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-003; E-004

Requirement ids: FR-009; FR-010; FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Component name: Security Alert and Audit Module

Responsibility: The system shall generate security alerts. The system shall maintain audit logs of finding updates and administrator actions. The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Centralized Dashboard

Detail: Proposed logical owner for FR-011, FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-011; FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Component name: Centralized Dashboard

Responsibility: The system shall provide a centralized dashboard. The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Identity Integration Service

Detail: Proposed logical owner for FR-012, FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-012; FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Component name: Identity Integration Service

Responsibility: The system shall integrate with existing university identity services. The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-002, FR-003, FR-004, FR-005, FR-006.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006

From component: Actor: Security analysts

To component: Finding Management Module

Data or action: The system shall allow Security analysts to review findings. The system shall allow Security analysts to assign remediation tasks to authorized departmental administrators. The system shall allow Security analysts to record investigation notes. The system shall allow Security analysts to track remediation status. The system shall allow Security analysts to close findings after reviewing submitted evidence.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Finding Management Module

To component: Access Control Engine

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Finding Management Module

To component: Security Alert and Audit Module

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The

system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Finding Management Module

To component: Centralized Dashboard

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Finding Management Module

To component: Identity Integration Service

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Access Control Engine

To component: Finding Management Module

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Access Control Engine

To component: Security Alert and Audit Module

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Access Control Engine

To component: Centralized Dashboard

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Access Control Engine

To component: Identity Integration Service

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Security Alert and Audit Module

To component: Finding Management Module

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Security Alert and Audit Module

To component: Access Control Engine

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Security Alert and Audit Module

To component: Centralized Dashboard

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Security Alert and Audit Module

To component: Identity Integration Service

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Centralized Dashboard

To component: Finding Management Module

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Centralized Dashboard

To component: Access Control Engine

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Centralized Dashboard

To component: Security Alert and Audit Module

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Centralized Dashboard

To component: Identity Integration Service

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Identity Integration Service

To component: Finding Management Module

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Identity Integration Service

To component: Access Control Engine

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Identity Integration Service

To component: Security Alert and Audit Module

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-013, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Identity Integration Service

To component: Centralized Dashboard

Data or action: The system shall support multiple campus locations. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-002, FR-003, FR-004, FR-005, FR-006.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006

Untrusted component: Actor: Security analysts

Trusted component: Finding Management Module

Enforcement: The system shall allow Security analysts to review findings. The system shall allow Security analysts to assign remediation tasks to authorized departmental administrators. The system shall allow Security analysts to record investigation notes. The system shall allow Security analysts to track remediation status. The system shall allow Security analysts to close findings after reviewing submitted evidence.

9. Threats

Threats

Id: THR-001

Asset: User identities and authenticated access

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Impact: Unauthorized use of a legitimate identity could expose resources or alter system state.

Control requirement ids: FR-011; FR-012; FR-013; NFR-002; NFR-003; NFR-004; NFR-005

Evidence ids: E-004

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-002

Asset: Role and permission boundaries

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: A principal exploits an incorrect role assignment or permission check to perform operations outside the source-defined authorization boundary.

Impact: A lower-privilege user could access restricted functions or sensitive resources.

Control requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010

Evidence ids: E-002; E-003

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-003

Asset: Monitoring and security alerts

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Impact: Relevant suspicious activity could remain undetected or receive delayed investigation.

Control requirement ids: FR-007; FR-008; FR-009; FR-010

Evidence ids: E-003

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-004

Asset: Integration trust paths

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Impact: Untrusted integration data could cause incorrect decisions or hide security events.

Control requirement ids: FR-011; FR-012; FR-013; NFR-002; NFR-003; NFR-004; NFR-005

Evidence ids: E-004

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-005

Asset: Security and administrative audit evidence

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Impact: Investigators could lose reliable evidence of the events required by the brief.

Control requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010

Evidence ids: E-002; E-003

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

Risks

Id: RISK-001

Description: Derived risk affecting User identities and authenticated access: Unauthorized use of a legitimate identity could expose resources or alter system state.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Mitigation: Review impersonation and compromised-identity paths against the stated access duties; confirm any additional identity or session protections with the stakeholder. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-011; FR-012; FR-013; NFR-002; NFR-003; NFR-004; NFR-005

Evidence ids: E-004

Owner: TBD

Id: RISK-002

Description: Derived risk affecting Role and permission boundaries: A lower-privilege user could access restricted functions or sensitive resources.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A principal exploits an incorrect role assignment or permission check to perform operations outside the source-defined authorization boundary.

Mitigation: Test permitted and denied role/resource pairs against the linked authorization duties; review ambiguous permission assignments with the resource owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010

Evidence ids: E-002; E-003

Owner: TBD

Id: RISK-003

Description: Derived risk affecting Monitoring and security alerts: Relevant suspicious activity could remain undetected or receive delayed investigation.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Mitigation: Evaluate the stated detection duties with representative positive and negative events; confirm undecided thresholds, alert priorities and investigation responsibilities. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-007; FR-008; FR-009; FR-010

Evidence ids: E-003

Owner: TBD

Id: RISK-004

Description: Derived risk affecting Integration trust paths: Untrusted integration data could cause incorrect decisions or hide security events.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Mitigation: Review trust assumptions at the specified integration boundaries with missing or altered input fixtures; confirm the required failure handling with the owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-011; FR-012; FR-013; NFR-002; NFR-003; NFR-004; NFR-005

Evidence ids: E-004

Owner: TBD

Id: RISK-005

Description: Derived risk affecting Security and administrative audit evidence: Investigators could lose reliable evidence of the events required by the brief.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Mitigation: Compare source-required events with the resulting audit evidence; assess missing and altered-record cases and confirm unprovided protection or retention decisions. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010

Evidence ids: E-002; E-003

Owner: TBD

11. User stories and use cases

User stories and use cases

Id: UC-001

Actor: Security analysts

Goal: The system shall allow Security analysts to review findings.

Story: As Security analysts, I need the following behavior: The system shall allow Security analysts to review findings.

Preconditions

1. Security analysts review findings named in the request

Steps

1. Security analysts initiate the review action

Success outcome: The requested findings are returned for examination; an external test capture records the response

Requirement ids: FR-002

Evidence ids: E-002

Id: UC-002

Actor: Security analysts

Goal: The system shall allow Security analysts to assign remediation tasks to authorized departmental administrators.

Story: As Security analysts, I need the following behavior: The system shall allow Security analysts to assign remediation tasks to authorized departmental administrators.

Preconditions

1. Security analysts assign remediation tasks to authorized departmental administrators

Steps

1. Security analysts initiate the assignment operation

Success outcome: The permission or access state for authorized departmental administrators is observed as active after Security analysts initiate the assignment operation.

Requirement ids: FR-003

Evidence ids: E-002

Id: UC-003

Actor: Security analysts

Goal: The system shall allow Security analysts to record investigation notes.

Story: As Security analysts, I need the following behavior: The system shall allow Security analysts to record investigation notes.

Preconditions

1. Security analysts perform a controlled instance of recording investigation notes

Steps

1. Security analysts initiate the record operation

Success outcome: The requested investigation note record is available for inspection; an external test capture records the inspection record

Requirement ids: FR-004

Evidence ids: E-002

Id: UC-004

Actor: Security analysts

Goal: The system shall allow Security analysts to track remediation status.

Story: As Security analysts, I need the following behavior: The system shall allow Security analysts to track remediation status.

Preconditions

1. Security analysts track remediation status for a specific finding

Steps

1. Security analysts initiate the tracking operation

Success outcome: The named history of remediation status is available for examination; an external test capture records the resulting state

Requirement ids: FR-005

Evidence ids: E-002

Id: UC-005

Actor: Security analysts

Goal: The system shall allow Security analysts to close findings after reviewing submitted evidence.

Story: As Security analysts, I need the following behavior: The system shall allow Security analysts to close findings after reviewing submitted evidence.

Preconditions

1. Security analysts close findings after reviewing submitted evidence

Steps

1. Security analysts initiate the closure action

Success outcome: The finding is marked as closed; an external test capture records the updated status response

Requirement ids: FR-006

Evidence ids: E-002

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: the system receives an approved security assessment report for a server
2. When: the system manages the findings from that report

Expected result: Expected: the system returns a structured finding list with severity, location, and recommended remediation steps

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-002

Requirement ids: FR-002

Procedure

1. Given: Security analysts review findings named in the request
2. When: Security analysts initiate the review action

Expected result: Expected: The requested findings are returned for examination; an external test capture records the response

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-003

Requirement ids: FR-003

Procedure

1. Given: Security analysts assign remediation tasks to authorized departmental administrators
2. When: Security analysts initiate the assignment operation

Expected result: Expected: The permission or access state for authorized departmental administrators is observed as active after Security analysts initiate the assignment operation.

Evidence to record: Evidence: External test capture of the resulting state

Test type: inspection

Id: TEST-004

Requirement ids: FR-004

Procedure

1. Given: Security analysts perform a controlled instance of recording investigation notes
2. When: Security analysts initiate the record operation

Expected result: Expected: The requested investigation note record is available for inspection; an external test capture records the inspection record

Evidence to record: Evidence: External test capture of the inspection record

Test type: analysis

Id: TEST-005

Requirement ids: FR-005

Procedure

1. Given: Security analysts track remediation status for a specific finding
2. When: Security analysts initiate the tracking operation

Expected result: Expected: The named history of remediation status is available for examination; an external test capture records the resulting state

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-006

Requirement ids: FR-006

Procedure

1. Given: Security analysts close findings after reviewing submitted evidence
2. When: Security analysts initiate the closure action

Expected result: Expected: The finding is marked as closed; an external test capture records the updated status response

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-007

Requirement ids: FR-007

Procedure

1. Given: system restricts access to resources based on user roles
2. When: the system enforces role-based access control during resource retrieval

Expected result: Expected: The system enforces role-based access control during resource retrieval, and access to resources is restricted according to user roles.

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-008

Requirement ids: FR-008

Procedure

1. Given: system monitors changes to findings or remediation records
2. When: an unauthorized modification is attempted on a finding or remediation record

Expected result: Expected: the system detects and logs the change as unauthorized

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-009

Requirement ids: FR-009

Procedure

1. Given: system generates security alerts under defined conditions
2. When: a predefined alert condition is initiated by a threat or anomaly event

Expected result: Expected: a security alert exists in the designated alerting system with required fields populated

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-010

Requirement ids: FR-010

Procedure

1. Given: system maintains audit logs for finding updates and administrator actions
2. When: an update to a finding or an administrative action is performed

Expected result: Expected: The system maintains audit logs for finding updates and administrator actions, as evidenced by external test capture of the inspection record.

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Id: TEST-011

Requirement ids: FR-011

Procedure

1. Given: a controlled user opens the source-named dashboard
2. When: the system provides a centralized dashboard

Expected result: Expected: the dashboard is displayed or available for use

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-012

Requirement ids: FR-012

Procedure

1. Given: a controlled user initiates an identity service exchange with the university identity services
2. When: the system integrates with existing university identity services

Expected result: Expected: interoperation with the external service is observed

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-013

Requirement ids: FR-013

Procedure

1. Given: a controlled user initiates use from each stated campus location
2. When: the system supports multiple campus locations

Expected result: Expected: use succeeds at each location

Evidence to record: Evidence: External test capture of the inspection record

Test type: positive

Id: TEST-014

Requirement ids: NFR-001

Procedure

1. Given: A draft acceptance plan for this quality goal: the system shall be secure against known attack vectors and maintain confidentiality, integrity, and availability
2. When: An analyst inspects or reviews that acceptance plan to determine whether measurable targets are defined

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Id: TEST-015

Requirement ids: NFR-002

Procedure

1. Given: The system shall remain secure during vulnerability remediation operations.
2. When: An analyst inspects or reviews that acceptance plan.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed.

Evidence to record: Evidence: External test capture of the resulting response

Test type: inspection

Id: TEST-016

Requirement ids: NFR-003

Procedure

1. Given: The system shall remain reliable during vulnerability remediation operations.
2. When: An analyst inspects or reviews that acceptance plan.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed.

Evidence to record: Evidence: External test capture of the resulting state

Test type: inspection

Id: TEST-017

Requirement ids: NFR-004

Procedure

1. Given: The system shall remain scalable during vulnerability remediation operations.
2. When: An analyst inspects or reviews that acceptance plan.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed.

Evidence to record: Evidence: External test capture of the inspection record

Test type: analysis

Id: TEST-018

Requirement ids: NFR-005

Procedure

1. Given: The system shall remain manageable during vulnerability remediation operations.
2. When: An analyst inspects or reviews that acceptance plan.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed.

Evidence to record: Evidence: External test capture of the resulting state

Test type: inspection

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Log

Detail: A collection of event records. This term does not itself require additional logging fields, events or access permissions.

Status: proposed

Evidence ids: E-003

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-003

14. SDLC recommendation and workflow

Selected model: Incremental

Justification: Derived operational_validation (+6 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003) Derived deployment_scope (+12 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004) Derived integration_assurance (+8 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.
2. Proposed suitability recommendation; confirm delivery constraints and team capacity before adopting this lifecycle. Missing factors are not assumed.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: The system shall manage findings from approved security assessment reports for servers, applications, endpoints, and network devices.; FR-002: The system shall allow Security analysts to review findings.; FR-003: The system shall allow Security analysts to assign remediation tasks to authorized departmental administrators.; FR-004: The system shall allow Security analysts to record investigation notes.; FR-005: The system shall allow Security analysts to track remediation status.; FR-006: The system shall allow Security analysts to close findings after reviewing submitted evidence.; FR-007: The system shall restrict access according to user roles.; FR-008: The system shall detect unauthorized changes to findings or remediation records.; FR-009: The system shall generate security alerts.; FR-010: The system shall maintain audit logs of finding updates and administrator actions.; FR-011: The system shall provide a centralized dashboard.; FR-012: The system shall integrate with existing university identity services.; FR-013: The system shall support multiple campus locations.
2. Review and baseline the source-defined work group: NFR-001: The system shall be secure.; NFR-002: The system shall remain secure.; NFR-003: The system shall remain reliable.; NFR-004: The system shall remain scalable.; NFR-005: The system shall remain manageable.
3. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline
2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review implementation and acceptance evidence against FR-007: The system shall restrict access according to user roles.
2. Review implementation and acceptance evidence against FR-009: The system shall generate security alerts.
3. Review implementation and acceptance evidence against NFR-001: The system shall be secure.
4. Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; FR-012; FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Name: Incremental delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: The system shall manage findings from approved security assessment reports for servers, applications, endpoints, and network devices.; FR-002: The system shall allow Security analysts to review findings.; FR-003: The system shall allow Security analysts to assign remediation tasks to authorized departmental administrators.; FR-004: The system shall allow Security analysts to record investigation notes.; FR-005: The system shall allow Security analysts to track remediation status.; FR-006: The system shall allow Security analysts to close findings after reviewing submitted evidence.; FR-007: The system shall restrict access according to user roles.; FR-008: The system shall detect unauthorized changes to findings or remediation records.; FR-009: The system shall generate security alerts.; FR-010: The system shall maintain audit logs of finding updates and administrator actions.; FR-011: The system shall provide a centralized dashboard.; FR-012: The system shall integrate with existing university identity services.; FR-013: The system shall support multiple campus locations.
3. Implement the source-defined work group: NFR-001: The system shall be secure.; NFR-002: The system shall remain secure.; NFR-003: The system shall remain reliable.; NFR-004: The system shall remain scalable.; NFR-005: The system shall remain manageable.
4. In an authorized isolated test environment, execute TEST-001: Given: the system receives an approved security assessment report for a server When: the system manages the findings from that report Expected: the system returns a structured finding list with severity, location, and recommended remediation steps
5. In an authorized isolated test environment, execute TEST-002: Given: Security analysts review findings named in the request When: Security analysts initiate the review action Expected: The requested findings are returned for examination; an external test capture records the response
6. In an authorized isolated test environment, execute TEST-003: Given: Security analysts assign remediation tasks to authorized departmental administrators When: Security analysts initiate the assignment operation Expected: The permission or access state for authorized departmental administrators is observed as active after Security analysts initiate the assignment operation.
7. In an authorized isolated test environment, execute TEST-004: Given: Security analysts perform a controlled instance of recording investigation notes When: Security analysts initiate the record

operation Expected: The requested investigation note record is available for inspection; anexternal test capture records the inspection record

8. In an authorized isolated test environment, execute TEST-005: Given: Security analysts track remediation status for a specific finding When: Security analysts initiate the tracking operation Expected: The named history of remediation status is available for examination; an external test capture records the resulting state
9. In an authorized isolated test environment, execute TEST-006: Given: Security analysts close findings after reviewing submitted evidence When: Security analysts initiate the closure action Expected: The finding is marked as closed; an external test capture records the updated status response
10. In an authorized isolated test environment, execute TEST-007: Given: system restricts access to resources based on user roles When: the system enforces role-based access control during resource retrieval Expected: The system enforces role-based access control during resource retrieval, and access to resources is restricted according to user roles.
11. In an authorized isolated test environment, execute TEST-008: Given: system monitors changes to findings or remediation records When: an unauthorized modification is attempted on a finding or remediation record Expected: the system detects and logs the change as unauthorized
12. In an authorized isolated test environment, execute TEST-009: Given: system generates security alerts under defined conditions When: a predefined alert condition is initiated by a threat or anomaly event Expected: a security alert exists in the designated alerting system with required fields populated
13. In an authorized isolated test environment, execute TEST-010: Given: system maintains audit logs for finding updates and administrator actions When: an update to a finding or an administrative action is performed Expected: The system maintains audit logs for finding updates and administrator actions, as evidenced by external test capture of the inspection record.
14. In an authorized isolated test environment, execute TEST-011: Given: a controlled user opens the source-named dashboard When: the system provides a centralized dashboard Expected: the dashboard is displayed or available for use
15. In an authorized isolated test environment, execute TEST-012: Given: a controlled user initiates an identity service exchange with the university identity services When: the system integrates with existing university identity services Expected: interoperation with the external service is observed
16. In an authorized isolated test environment, execute TEST-013: Given: a controlled user initiates use from each stated campus location When: the system supports multiple campus locations Expected: use succeeds at each location
17. In an authorized isolated test environment, execute TEST-014: Given: A draft acceptance plan for this quality goal: the system shall be secure against known attack vectors and maintain confidentiality, integrity, and availability When: An analyst inspects or reviews that acceptance plan to determine whether measurable targets are defined Expected: The plan's measurable targets remain TBD or unconfirmed
18. In an authorized isolated test environment, execute TEST-015: Given: The system shall remain secure during vulnerability remediation operations. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
19. In an authorized isolated test environment, execute TEST-016: Given: The system shall remain reliable during vulnerability remediation operations. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
20. In an authorized isolated test environment, execute TEST-017: Given: The system shall remain scalable during vulnerability remediation operations. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.

21. In an authorized isolated test environment, execute TEST-018: Given: The system shall remain manageable during vulnerability remediation operations. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
22. Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
23. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review implementation and acceptance evidence against FR-007: The system shall restrict access according to user roles.
2. Review implementation and acceptance evidence against FR-009: The system shall generate security alerts.
3. Review implementation and acceptance evidence against NFR-001: The system shall be secure.
4. Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; FR-012; FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review implementation and acceptance evidence against FR-007: The system shall restrict access according to user roles.
2. Review implementation and acceptance evidence against FR-009: The system shall generate security alerts.
3. Review implementation and acceptance evidence against NFR-001: The system shall be secure.
4. Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; FR-012; FR-013; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Unknown factors: budget; delivery; failure_consequences; feedback; legacy; regulation; schedule; stability; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-2.0

Generation profile: sequential-sdlc-generation-2.1

Factors

Name: security_assurance

Value: integrated security review

Evidence ids: E-002

Basis: derived

Rationale: Security decisions benefit from security review and tests throughout delivery.

Name: operational_validation

Value: monitoring feedback

Evidence ids: E-003

Basis: derived

Rationale: Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency.

Name: deployment_scope

Value: staged rollout opportunity

Evidence ids: E-004

Basis: derived

Rationale: Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed.

Name: integration_assurance

Value: integration validation

Evidence ids: E-004

Basis: derived

Rationale: Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified.

Ranked candidates

Model: Incremental

Score: 66

Reasons: Derived operational_validation (+6 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003); Derived deployment_scope (+12 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004); Derived integration_assurance (+8 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

Model: Hybrid

Score: 65

Reasons: Derived security_assurance (+8 points): Security decisions benefit from security review and tests throughout delivery. (E-002); Derived operational_validation (+5 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003); Derived deployment_scope (+8 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004); Derived integration_assurance (+4 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

Model: DevSecOps

Score: 63

Reasons: Derived security_assurance (+12 points): Security decisions benefit from security review and tests throughout delivery. (E-002); Derived operational_validation (+8 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003); Derived deployment_scope (+3 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004)

Model: Spiral

Score: 51

Reasons: Derived security_assurance (+5 points): Security decisions benefit from security review and tests throughout delivery. (E-002); Derived integration_assurance (+6 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

Model: Agile

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 40

Reasons: No differentiating input evidence

Evidence register

Id: E-001

Source: description

Quote: Design a secure vulnerability-remediation tracking system for a university that manages findings from approved security assessment reports for servers, applications, endpoints, and network devices.

Id: E-002

Source: description

Quote: Security analysts should be able to review findings, assign remediation tasks to authorized departmental administrators, record investigation notes, track remediation status, and close findings after reviewing submitted evidence.

Id: E-003

Source: description

Quote: The system should restrict access according to user roles, detect unauthorized changes to findings or remediation records, generate security alerts, and maintain audit logs of finding updates and administrator actions.

Id: E-004

Source: description

Quote: The platform should provide a centralized dashboard, integrate with existing university identity services, support multiple campus locations, and remain secure, reliable, scalable, and manageable.

Source references

kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.7394

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:8 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.7289

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-ecd06e0e85ebe229:17 - 03_OWASP_CheatSheets/038_OWASP_Authorization_Cheat_Sheet.md

Source document id: kb-ecd06e0e85ebe229

Chunk index: 17

Page or section: text chunk 18

Relevance score: 0.7156

Sha256: e78a10eb9759a1fc7593ba1fc89a05b017e7e795ac16ad5496bdba86138d891f

kb-ecd06e0e85ebe229:16 - 03_OWASP_CheatSheets/038_OWASP_Authorization_Cheat_Sheet.md

Source document id: kb-ecd06e0e85ebe229

Chunk index: 16

Page or section: text chunk 17

Relevance score: 0.7241

Sha256: e78a10eb9759a1fc7593ba1fc89a05b017e7e795ac16ad5496bdba86138d891f

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Non-functional requirements: cybersrs-qwen3-4b-ft:latest

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: cybersrs-qwen3-4b-ft:latest

Prompt version: sequential-1.37

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-001	TEST-001, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-002	E-002	UC-001, TEST-002, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-003	E-002	UC-002, TEST-003, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-004	E-002	UC-003, TEST-004, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-005	E-002	UC-004, TEST-005, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-006	E-002	UC-005, TEST-006, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-007	E-003	TEST-007, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-008	E-003	TEST-008, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-009	E-003	TEST-009, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-010	E-003	TEST-010, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-011	E-004	TEST-011, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-012	E-004	TEST-012, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-013	E-004	TEST-013, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-001	E-001	TEST-014, Requirements and verification planning, Incremental delivery cycle, Release and handoff

Requirement	Evidence / sources	Stories / tests / SDLC
NFR-002	E-004	TEST-015, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-003	E-004	TEST-016, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-004	E-004	TEST-017, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-005	E-004	TEST-018, Requirements and verification planning, Incremental delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.37